Chandan Kumar Nayak

Bhagabanpur, Ganjam, Odisha, 761151

9777064700 chandankunayak2003@gmail.com Linkedin GitHub

Professional Summary

AI/ML Engineer with hands-on experience in building LLM-powered applications, NLP pipelines, and computer vision systems. Skilled in Generative AI, FastAPI-based model deployment, and scalable microservices architecture. Proven ability to design production-ready AI solutions including semantic search, face recognition systems, and workflow automation using OpenAI APIs.

Education

National Institute of Science and Technology (Autonomous) — **Aug. 2020 – April. 2024**
Bachelor of Technology in Information and Technology — *Berhampur, Odisha*

Om Bhurbhuvah Svah Residential Higher Secondary School — **May. 2018 – March. 2020**
Intermediate — *Berhampur, Odisha*

Ashramika High School — **April. 2017 – March. 2018**
Matriculation — *Asurabandha, Odisha*

Technical Skills

Languages: SQL, Python, Java, Angualr, JavaScript

Developer Tools: VS Code, Jupyter Notebook, Spyder, PyCharm, Power BI, Excel

Technologies/Frameworks: Machine Learning Algorithms(Linear Regression, Polynomial Regression, Logistic Regression, SVM, KNN, Random Forest, Decision Tree, Naive Bayes algorithm), Deep Learning, computer vision, Artificial Intelligence(AI), Generative AI, Large Language Model(LLM), Hugging face platform, Prompt Engineering, Stream-lit

Experience

Dhanush Softech — **June 2025 – Present**
AI/ML Engineer — *Hyderabad, Telangana*

- **OUK (Open University of Kenya)** – Architected LLM-powered automation workflows using OpenAI APIs and n8n orchestration to streamline academic processes including AI-driven content generation, document processing, and intelligent task routing.
- Designed transformer-based NLP pipelines leveraging Hugging Face models for text embeddings, semantic search, and contextual response generation; deployed scalable inference services using FastAPI for real-time AI operations.
- **KEMIS – AI-Based Attendance Management System** – Developed a computer vision pipeline using ArcFace for high-accuracy face recognition and biometric attendance tracking.
- Implemented facial embedding extraction, cosine similarity matching, and real-time inference optimization; exposed ML models via production-grade REST APIs ensuring scalability and low-latency response.
- **SARADHYAM** – Engineered a full-stack enterprise platform using Angular, FastAPI, and PostgreSQL, designing modular frontend architecture and scalable RESTful microservices.

- Designed normalized database schemas, optimized SQL queries and indexing strategies, and implemented authentication/authorization mechanisms ensuring secure and high-performance backend operations.
- Integrated backend APIs with dynamic Angular components, implemented state management, form validation, and responsive UI design to enhance usability and maintainability.
- **MMU Chhattisgarh** – Developed dynamic gallery and content management modules using Angular, enabling efficient multimedia handling and structured academic content presentation.
- Optimized frontend rendering performance using lazy loading, reusable components, and modular architecture, improving page load efficiency and cross-device responsiveness.

Projects

AI-Generated-Imagery-with-Stable-Diffusion | *LLM,Hugging Face Diffusers:* **Dec. 2024 – Jan. 2025**

- Developed a Streamlit application for generating images from text prompts using the Stable Diffusion model, optimized with PyTorch for GPU acceleration.
- Integrated prompt engineering techniques and custom sliders for adjusting guidance scale and inference steps, improving user control over outputs.
- Technologies used : Stable Diffusion, PyTorch, Streamlit, Generative AI, Prompt Engineering
- Project Link - AI-Generated-Imagery-with-Stable-Diffusion

Knee-Osteoarthritis-using-CNN | *Deep Learning Algorithm* **Sept. 2023 – Feb. 2024**

- Developed a deep learning model to classify knee osteoarthritis severity from X-ray images using Convolutional Neural Networks (CNNs).
- Preprocessed medical imaging data, including grayscale conversion, resizing to 256x256 pixels, and normalization, for optimal model performance.
- Built a CNN model with multiple convolutional and pooling layers, achieving 90 percentage of test accuracy on the osteoarthritis classification task.
- Project Link - Knee-Osteoarthritis-using-CNN

Certification

Data-science/AI

- Data-science/AI certification from Magneq Software Hyderabad

NPTEL Cloud Computing

- Completed the Swayam NPTEL Cloud Computing course, mastering cloud architecture, virtualization, storage, and security, and gaining industry-relevant expertise to harness cloud technologies effectively.
- Certificate Link - NPTEL Cloud Computing

HackerRank(SQL Intermediate)

- Certified by HackerRank in SQL (Intermediate), showcasing adeptness in crafting complex queries, optimizing data workflows, and harnessing the power of relational databases with precision and efficiency
- Certificate Link - HackerRank(SQL Intermediate)